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Klasse: 6ACIF		Übung Nr.: 4_2
Lehrer: Prof. Spanring & Horny	Abgabedatum: 26.04.2025	Raum Nr.: Platz Nr.: Bewertung:



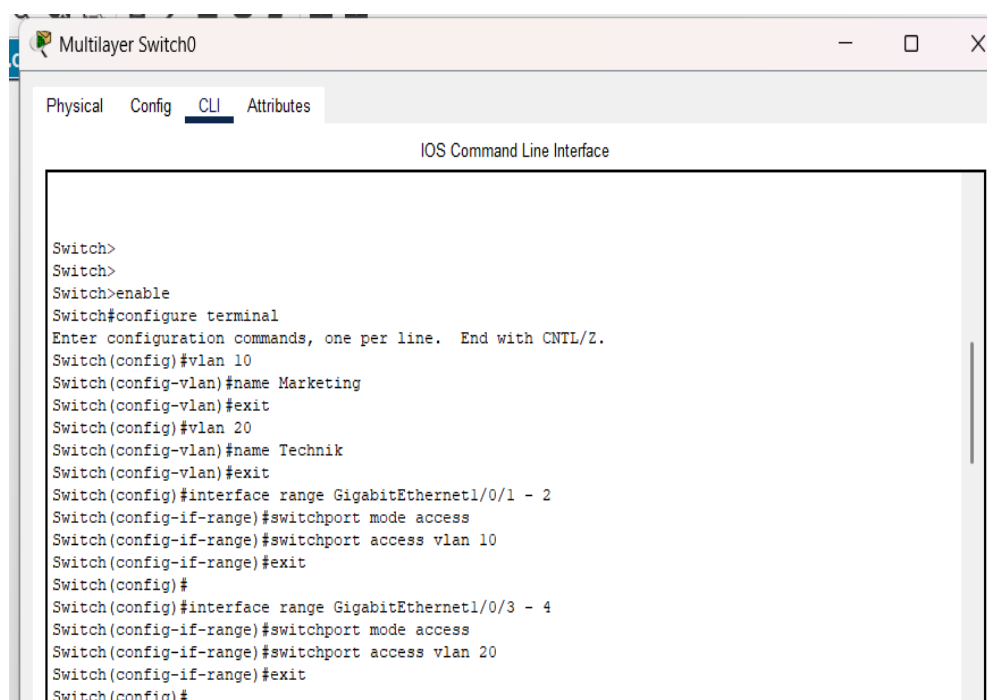
Laborübung Layer-3-Switch am Gerät (Teil 2)

1) Aufgabenstellung

- 4 PCs, verteilt auf 2 VLANs
- Layer 3 Switch (3650-24PS) für VLAN-Verwaltung
- VLAN 10 („Marketing“) → 192.168.10.0/24 → Ports 1 und 2
- VLAN 20 („Technik“) → 192.168.20.0/24 → Ports 3 und 4
- SVI auf dem Switch: IP-Adresse 10.0.0.1/24
- Router (2911) → IP-Adresse 10.0.0.2/24, DHCP-Server für beide VLANs
- DHCP-Adressen werden von Router zugewiesen
- Helper-Address wird auf dem Switch konfiguriert
- PC1 (VLAN 10) soll PC3 (VLAN 20) pingen können (Routing testen)

2) Switch-Konfiguration

VLANs anlegen und Ports zu VLANs zuweisen:



```
Multilayer Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch>
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Technik
Switch(config-vlan)#exit
Switch(config)#interface range GigabitEthernet1/0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range GigabitEthernet1/0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

Trunk-Port zum Router und IP-Routing aktivieren:

```
Switch(config)#  
Switch(config)#ip routing  
Switch(config)#interface GigabitEthernet1/0/5  
Switch(config-if)#switchport mode trunk  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
Switch(config)#
```

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☐ Top

SVI konfigurieren:

```
Switch(config)#interface vlan 10  
Switch(config-if)#ip address 192.168.8.1 255.255.255.0  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
%LINK-5-CHANGED: Interface Vlan10, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up  
  
Switch(config)#  
Switch(config)#interface vlan 20  
Switch(config-if)#ip address 192.168.80.1 255.255.255.0  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
%LINK-5-CHANGED: Interface Vlan20, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
```

Helper-Address setzen (DHCP-Relay):

```
Switch>  
Switch>enable  
Switch#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#  
Switch(config)#interface vlan 10  
Switch(config-if)#ip helper-address 10.0.0.2  
Switch(config-if)#exit  
Switch(config)#  
Switch(config)#interface vlan 20  
Switch(config-if)#ip helper-address 10.0.0.2  
Switch(config-if)#exit  
Switch(config)#
```

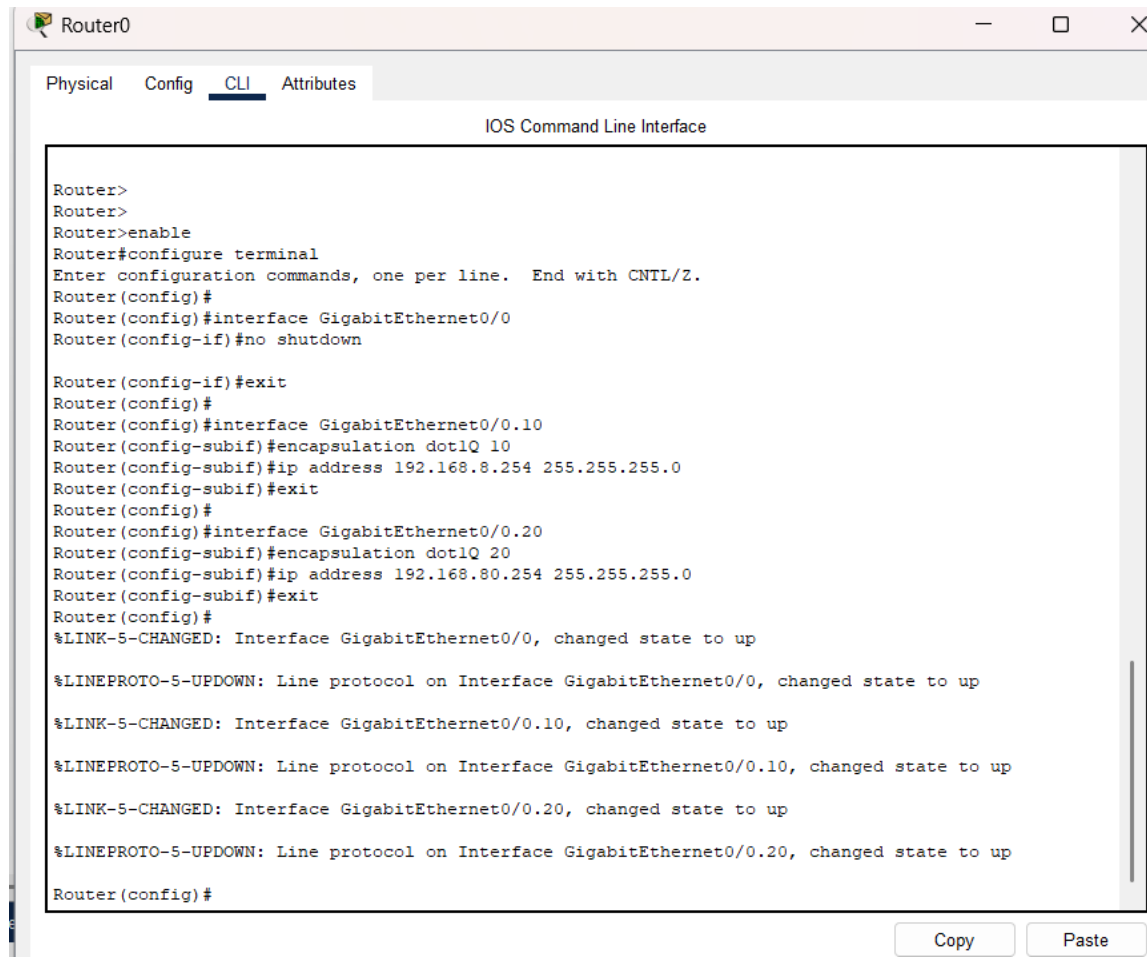
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3) Router-Konfiguration

Interface konfigurieren:



The screenshot shows the Router0 CLI interface with the following commands and output:

```
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.8.254 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.80.254 255.255.255.0
Router(config-subif)#exit
Router(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

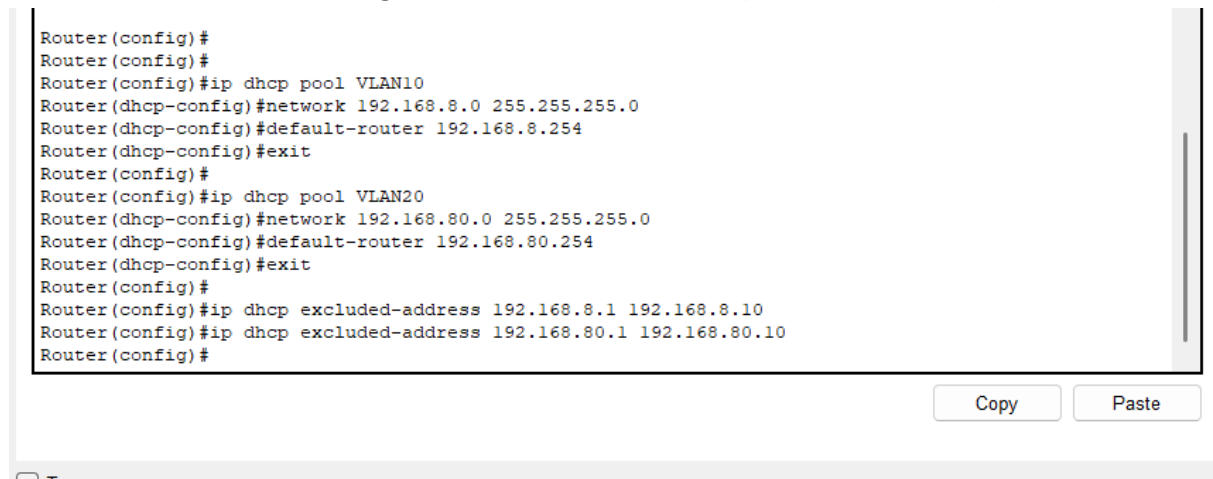
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config)#
```

Buttons: Copy, Paste

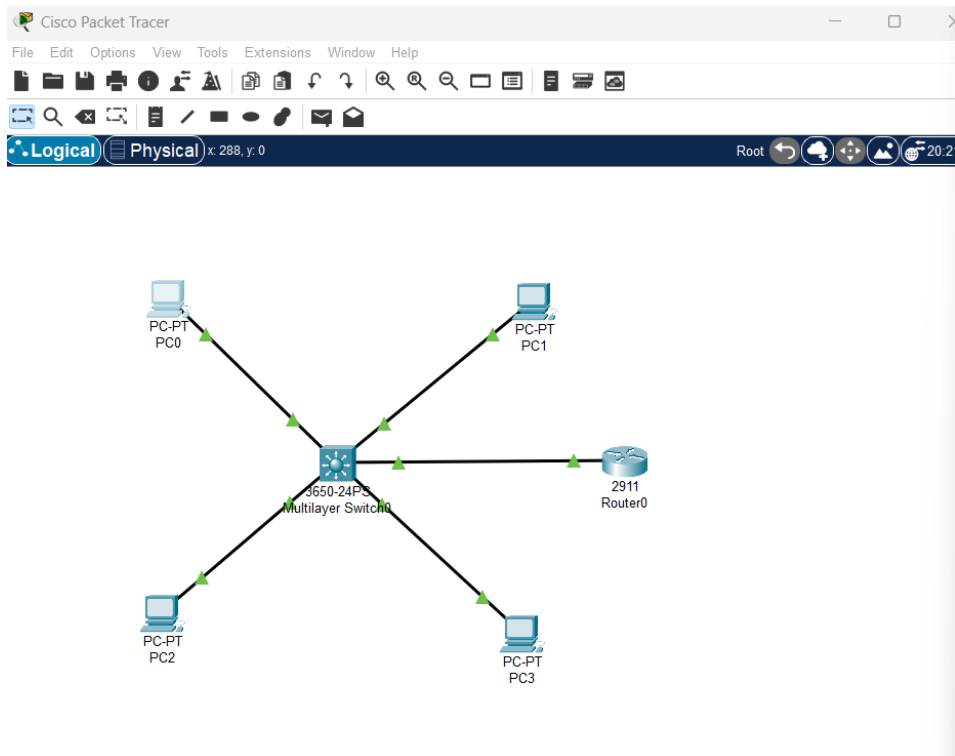
DHCP-Pools erstellen & Wichtige IPs vom DHCP ausnehmen (excluded-addresses):



The screenshot shows the Router0 CLI interface with the following commands and output:

```
Router(config)#
Router(config)#
Router(config)#ip dhcp pool VLAN10
Router(dhcp-config)#network 192.168.8.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.8.254
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool VLAN20
Router(dhcp-config)#network 192.168.80.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.80.254
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp excluded-address 192.168.8.1 192.168.8.10
Router(config)#ip dhcp excluded-address 192.168.80.1 192.168.80.10
Router(config)#
```

Buttons: Copy, Paste



4) Überprüfung (Tests)

Am PC1

(PC1 → PC3: Test, ob VLAN 10 zu VLAN 20 kommunizieren kann)

```
PC1
C:\>ping 192.168.80.11

Pinging 192.168.80.11 with 32 bytes of data:

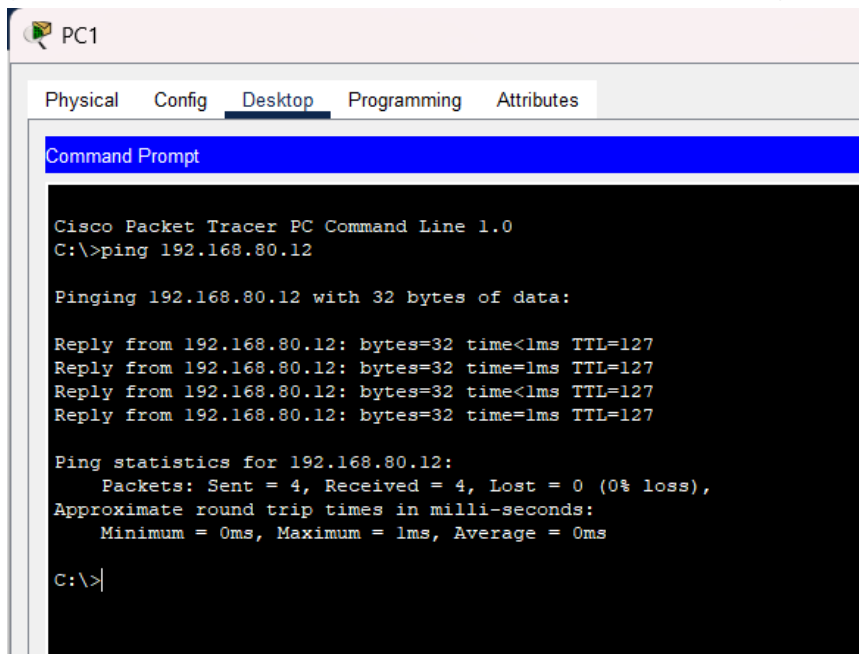
Reply from 192.168.80.11: bytes=32 time=10ms TTL=127
Reply from 192.168.80.11: bytes=32 time<1ms TTL=127
Reply from 192.168.80.11: bytes=32 time<1ms TTL=127
Reply from 192.168.80.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.80.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>|
```

Am PC1

(PC1 → PC2: Test, ob VLAN 10 zu VLAN 20 kommunizieren kann)



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.80.12

Pinging 192.168.80.12 with 32 bytes of data:

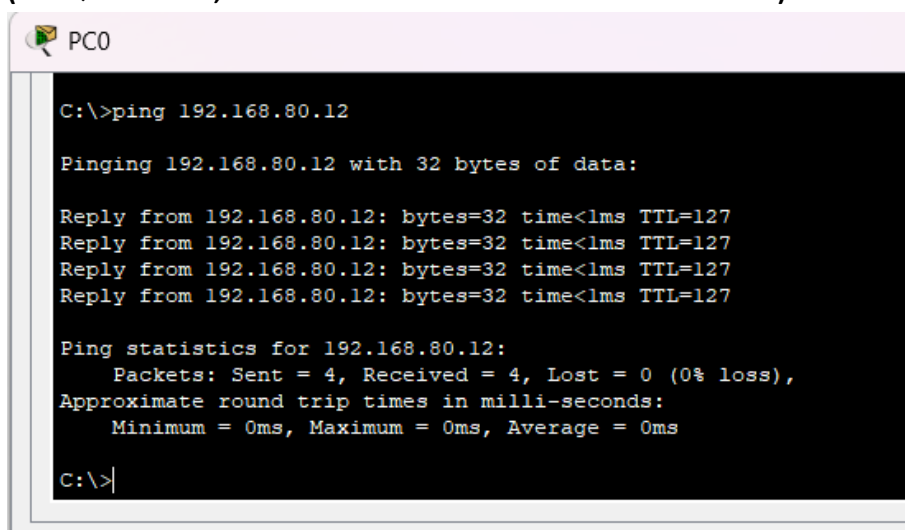
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.80.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Am PC0

(PC0 → PC2: Test, ob VLAN 10 zu VLAN 20 kommunizieren kann)



```
PC0

C:\>ping 192.168.80.12

Pinging 192.168.80.12 with 32 bytes of data:

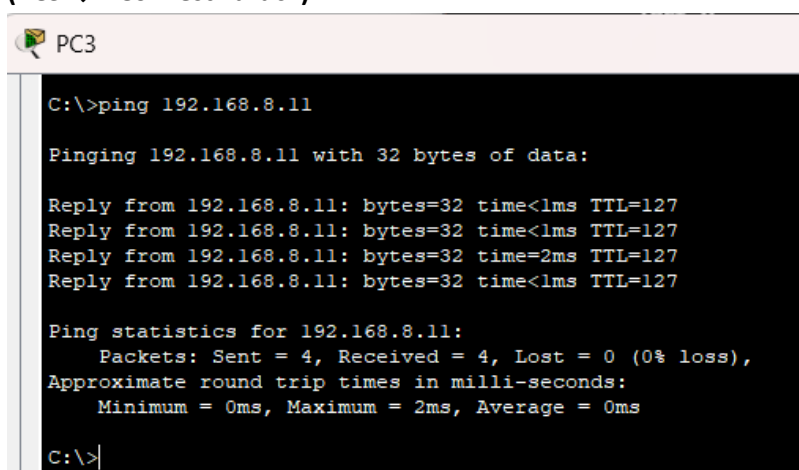
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127
Reply from 192.168.80.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.80.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Am PC3

(PC3 → PC0: Test zurück)



```
PC3

C:\>ping 192.168.8.11

Pinging 192.168.8.11 with 32 bytes of data:

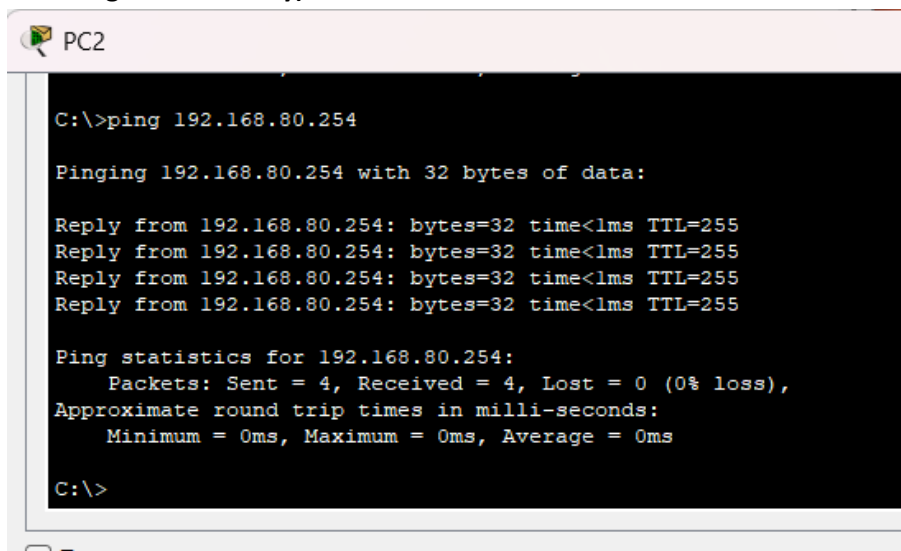
Reply from 192.168.8.11: bytes=32 time<1ms TTL=127
Reply from 192.168.8.11: bytes=32 time<1ms TTL=127
Reply from 192.168.8.11: bytes=32 time=2ms TTL=127
Reply from 192.168.8.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.8.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Am PC2

(Test zum eigenen Gateway)



```
C:\>ping 192.168.80.254

Pinging 192.168.80.254 with 32 bytes of data:

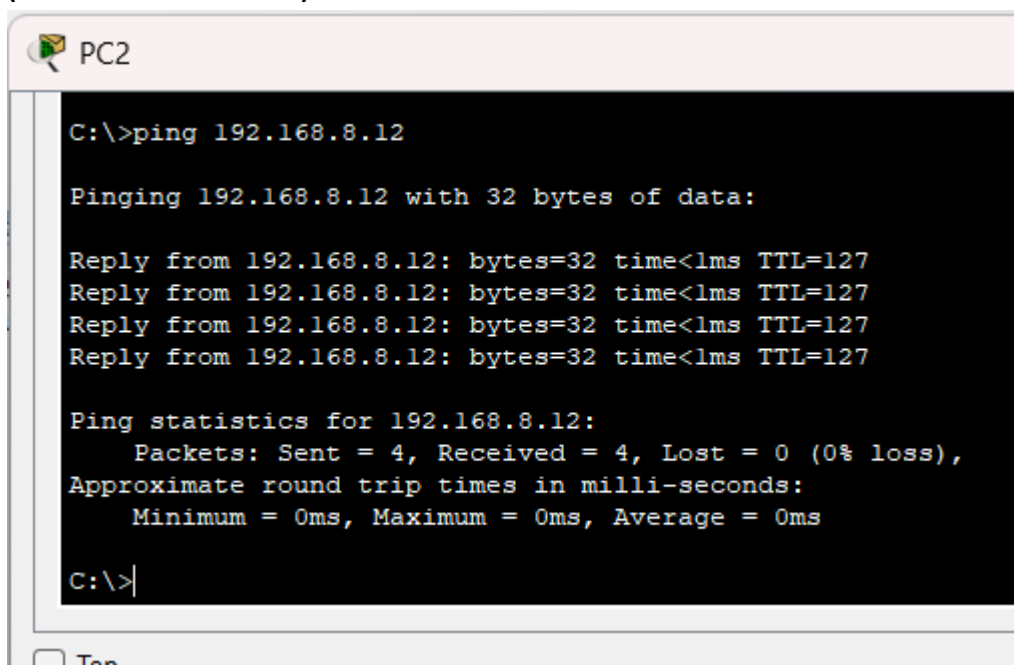
Reply from 192.168.80.254: bytes=32 time<1ms TTL=255
Reply from 192.168.80.254: bytes=32 time<1ms TTL=255
Reply from 192.168.80.254: bytes=32 time<1ms TTL=255
Reply from 192.168.80.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.80.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Am PC2

(PC2 → PC1: Test zurück)



```
C:\>ping 192.168.8.12

Pinging 192.168.8.12 with 32 bytes of data:

Reply from 192.168.8.12: bytes=32 time<1ms TTL=127
Reply from 192.168.8.12: bytes=32 time<1ms TTL=127
Reply from 192.168.8.12: bytes=32 time<1ms TTL=127
Reply from 192.168.8.12: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.8.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```